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AI-Powered Malaria Diagnosis System with Mobile, Web, and Chatbot Integration

Malaria Screener

FTL Ethiopia ML Bootcamp

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Concept note and Implementation Plan

Background

- As of 2023, malaria remains a significant global health challenge, with approximately 263 million new cases and nearly 600,000 deaths reported.
- The African continent bears the brunt of this burden, accounting for 94% of cases and 95% of malaria-related deaths.

Malaria is caused by five species of the Plasmodium parasite, with varying severity:

Plasmodium falciparum – The most dangerous and most common in Africa.

Objectives

Develop a cross-platform mobile app (Flutter) for AI-powered malaria screening, a Next.js dashboard for healthcare providers, a Django backend, and a Malaria-specific chatbot for instant support.

The system:

- Automate parasite detection in blood smears using AI-based models.
- Store and manage patient records.
- Provide a chatbot assistant for malaria-related queries.

Proposed Solution

Source: AI Generated Images

SDG Relation

Malaria is directly linked to the United Nations Sustainable Development Goals (SDGs),

particularly:

• SDG 3: Good Health and Well-being – Target 3.3 aims to end malaria by 2030.

"By 2030, end the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combat hepatitis, water-borne diseases and other communicable diseases"

Data

Data Collection

The project utilizes two primary sources of malaria blood smear images to ensure both diversity and clinical relevance:

NIH Malaria Dataset

- Samples: 27,408
- Type: Thin and thick blood smear images
- Origin: Publicly available dataset curated by the National Institutes of Health (NIH)

Exploratory Data Analysis and Feature Engineering

Dataset Overview:

- Source: Cell images from microscopy (Parasitized vs. Uninfected).
- Size: 27,558 images (13,779 Parasitized, 13,779 Uninfected).
- Split: 80% training (22,048), 20% validation (5,510), full test set (27,558).

Exploratory Data Analysis:

- Balanced classes ensure unbiased model training.
- Image resolution: Variable, resized to 128x128 pixels for consistency.
- Visual inspection: Parasitized cells show distinct ring-like structures; Uninfected cells are clearer.

Feature Engineering:

- Resizing: All images standardized to 128x128 pixels.
- Normalization: Pixel values scaled to [0, 1] (divided by 255).
- Data Augmentation:

Rotation (20°), zoom (20%), horizontal flips.

Enhances model robustness to variations in cell orientation and scale.

Model

Model Selection and Training

Custom CNN: Built from scratch for baseline performance (~3.3M parameters).

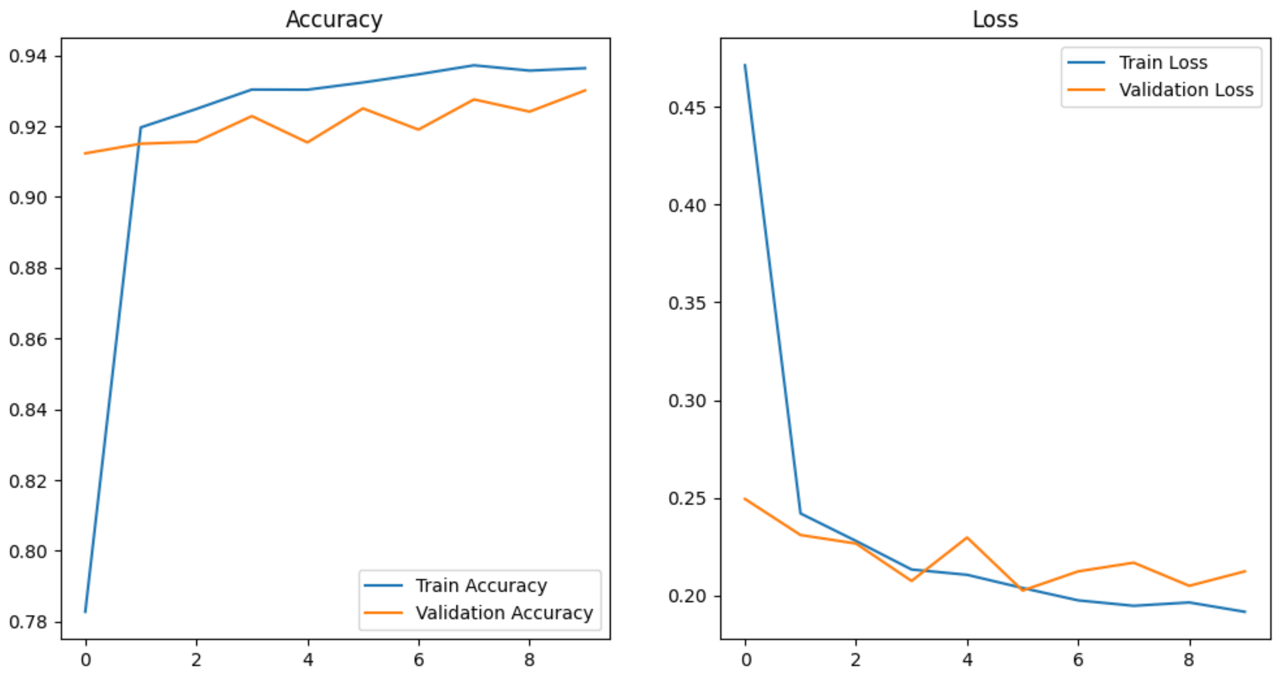
MobileNetV2: Pre-trained on ImageNet, optimized for efficiency (~2.42M parameters).

MobileNetV2 leverages transfer learning, suitable for mobile deployment.

Model Selection and Training

Custom CNN:

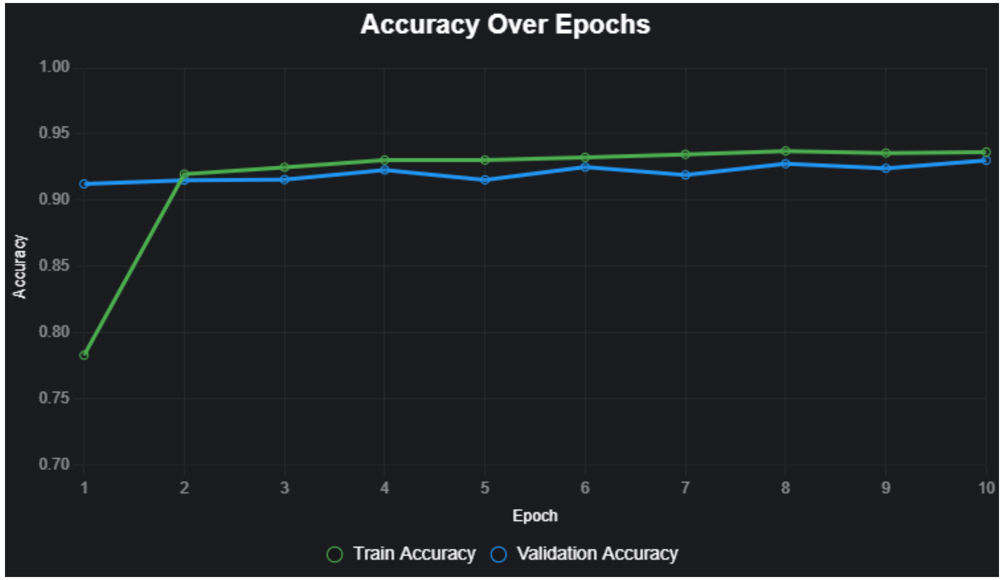
- loss: 0.2044
- accuracy: 0.9283
- Validation Accuracy: 0.93



Model Selection and Training

MobileNetV2 :

- loss: 0.2044
- accuracy: 0.94973
- Validation Accuracy: 0.95



Model Evaluation and Hyperparameter Tuning

Hyperparameter Tuning:

Custom CNN: Fixed settings (learning rate, batch size=32, Adam optimizer).

MobileNetV2:

- Learning rate: 1e-5 for fine-tuning (vs. default for initial training).
- Epochs: 5 (initial) + 10 (fine-tuning).
- Batch size: 32; dropout: 0.5 in head layers.

Tuning Strategy: Manual adjustment of learning rate for stability.

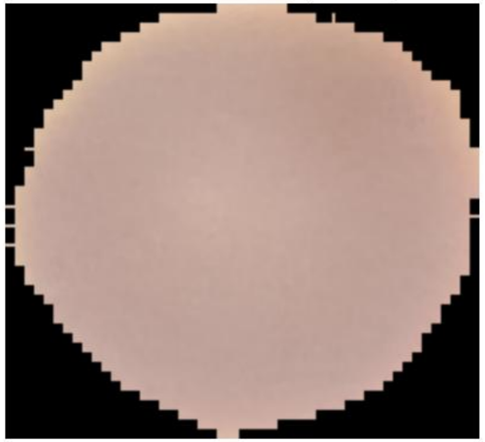
Model Refinement and Testing

MobileNetV2:

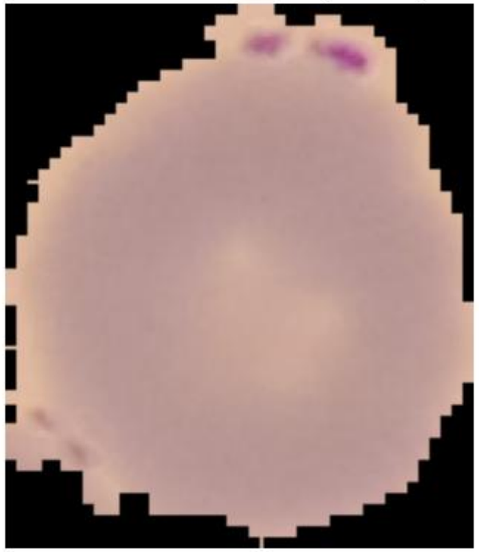
- Added custom head: Global average pooling, dense (128 units, ReLU), dropout (0.5), sigmoid output.
- Fine-tuning: Unfroze base layers, trained 10 epochs with low learning rate (1e-5).
- Improved validation accuracy from 92.49% (initial) to 94.97% (fine-tuned).

Model Refinement and Testing

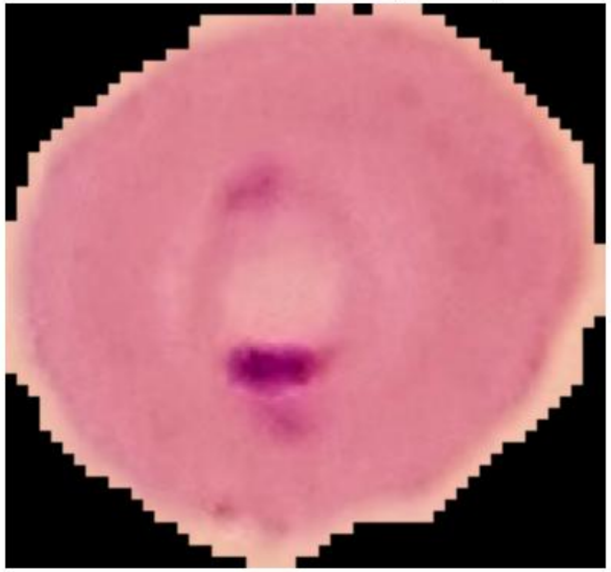
Predicted: Uninfected (99.68%)



Predicted: Parasitized (89.81%)



Predicted: Parasitized (99.99%)



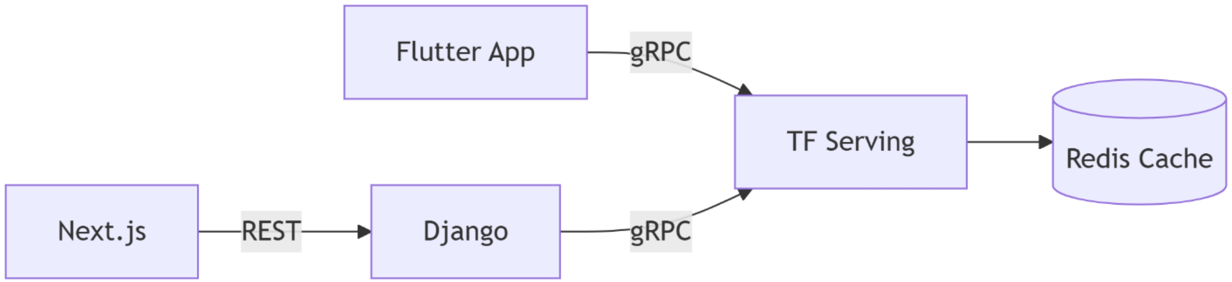
Predicted: Parasitized (100.00%)

Results

Evaluation Results

Metric	Custom CNN	MobileNetV2
Validation Accuracy	92.83% (5,510 images, 10 epochs)	94.97% (5,510 images, 15 epochs)
Test Accuracy	-	96% (27,558 images)
Precision	-	Parasitized: 0.99, Uninfected: 0.94
Recall	-	Parasitized: 0.94, Uninfected: 0.99
F1-Score	Not available	Parasitized: 0.96, Uninfected: 0.96
Evaluation Scope	Qualitative predictions (~171ms/image)	Full test set metrics, confusion matrix
Key Strength	Solid baseline performance	High accuracy, reliable for diagnosis
Limitation	No test set evaluation	Slight misclassifications (~4%)

Deployment



Deployment Architecture Overview

Data Flow:

- Flutter App → gRPC → TensorFlow Serving (for predictions).
- Next.js → REST → Django (for data retrieval).
- Django → gRPC → TensorFlow Serving.

Responses are cached in Redis for improved efficiency.

Key Benefits:

- Real-time Data Processing: Efficient communication through gRPC.
- Performance Optimization: Caching with Redis reduces response times.
- Scalability: Separation of mobile and web applications enhances manageability..

Future Work

1. We introduce a rapid, affordable smartphone application designed for malaria screening.
2. This app provides essential features with a user-friendly interface, enabling automatic screening of slides.
3. Management of images and metadata generated during the screening process, which can be utilized to further enhance the image analysis model.

Future Enhancements: detect multiple species, enable offline functionality, gather user feedback.

- Counting of infected red blood cells and parasites for P. falciparum malaria.
- Predicting Malaria Outbreaks

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Thank you!

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